

Research Paper

JWST/NIRCam F090W and F150W Surface Brightness Fluctuation Distances from GO-3055: TRGB-Anchored Calibrations and Internal Accuracy Tests

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Abstract

We measure surface brightness fluctuations (SBF) in JWST/NIRCam F090W and F150W images of 14 early-type galaxies from GO-3055, using a common measurement procedure and individual tip of the red giant branch (TRGB) distances. All galaxies enter the primary calibration. F150W does not require a colour correction in this sample: its adopted constant absolute magnitude is $-3.208 \pm 0.024_{\text{stat}} \pm 0.051_{\text{common}}$ mag, with intrinsic scatter 0.081 mag. Adding a linear colour term worsens the leave-one-out (LOO) root-mean-square (RMS) distance-modulus residual from 0.0925 to 0.0981 mag. In F090W, the relation $\bar{M}_{090,0} = -1.9617 + 1.8890[(F090W - F150W)_0 - 0.5703]$ mag reduces the LOO RMS from 0.0992 to 0.0684 mag and has intrinsic scatter 0.0489 mag. The median full absolute distance uncertainties are 0.0984 and 0.0752 mag for F150W and F090W, respectively, including the common 0.047-mag TRGB scale. We separate image-measurement, population, calibration and common-scale uncertainties, and compare distances with HST F110W and F160W results on a consistent error convention. These are internal cross-validations, not independent external accuracy tests. Correlated image noise, empirical PSF normalisation and population-dependent systematics remain limitations.

Keywords: Galaxy distances – Elliptical galaxies – Infrared photometry

1. Introduction

Surface brightness fluctuations measure the variance of unresolved stellar light per unit mean flux (Tonry & Schneider 1988). Their apparent amplitude decreases with distance, while their absolute magnitude depends on the stellar luminosity function. Near-infrared SBF benefits from luminous evolved stars but can require a stellar-population correction (Jensen et al. 2003, 2015, 2021). Colour is a potential predictor of this correction, not an ingredient of the image-based SBF measurement.

The TRGB–SBF project transfers geometrically anchored tip of the red giant branch (TRGB) distances to SBF. Jensen et al. (2025, Paper III) recalibrated Hubble Space Telescope (HST) F110W SBF using JWST TRGB distances and mean Virgo/Fornax anchors. Chazov et al. (2026, Paper IV) subsequently provided individual JWST TRGB distances. We use these anchors to measure and calibrate SBF directly in both JWST F090W and F150W. The two bands are processed with one methodology, allowing a controlled comparison of their colour response and internal distance precision.

We compare constant and linear calibrations before choosing the distance estimator. The primary experiment retains all 14 galaxies, without manual outlier removal. Technical derivations, complete residual-image montages and extended sensitivity tests are collected in the accompanying *Technical notes*, retained alongside this concise manuscript; the evidence needed to interpret the adopted distances is given here.

2. Data and common measurement method

2.1. Images, distance anchors and residuals

The input data are public GO-3055 James Webb Space Telescope (JWST) Near Infrared Camera (NIRCam) i2d mosaics in MJy sr^{−1}, with native pixel scale approximately 0.03123 arcsec. The sample comprises three Fornax galaxies, nine in the broad Virgo region and two other nearby group galaxies. The Virgo-region label is not strict cluster membership; in particular it includes NGC 4636 and foreground NGC 4697.

We use the individual Paper IV thin-slice TRGB moduli throughout the primary analysis. Thin slices were fixed before inspecting the present calibration residuals; Paper IV’s preferred thick-slice distances are retained as a sensitivity test. The SBF regions are circular annuli at 8.2–16.4 and 16.4–32.8 arcsec, following Paper III. The outer radius spans 1.65–3.17 kpc across the sample, so these are not matched physical apertures. The auxiliary elliptical-annulus test of Jensen et al. is not reproduced.

In each band we subtract a scalar sky, fit a smooth isophotal galaxy model with free ellipticity and position angle, and mask invalid pixels and compact sources. Model intensities and geometry are interpolated between isophotes; filled detector gaps are never used for SBF. For sky-subtracted image I_i and positive galaxy model M_i , define

$$Q_i = (I_i - M_i) / \sqrt{M_i}. \quad (1)$$

This removes the leading dependence of stellar variance on local galaxy light. A single median and scale are estimated with iterative clipped statistics over the complete catalogue-unmasked, finite, positive-model support, not separately in each annulus. Values of Q_i beyond median ± 3.5 scale are replaced by the boundary

values. This is *winsorisation*, not pixel rejection, and is an explicit departure from Jensen's procedure.

The full normalised, winsorised field is saved before selecting annuli. For binary annular window W_i , the Fourier input is

$$R_i = W_i \left(Q_i^{\text{win}} - \langle Q^{\text{win}} \rangle_W \right). \quad (2)$$

The mean is removed within each usable ring; masked pixels are absent rather than measurements of zero flux. Complete F150W and F090W montages in the Technical notes show the actual measurement support, black masked regions and the three annular boundaries.

2.2. Fourier amplitude and apparent SBF magnitude

The measured two-dimensional power spectrum is $P(k) = |\mathcal{F}[R]|^2 / N_{\text{use}}$, where N_{use} is the number of usable annular pixels. We average it into 79 radial bins defined by 80 edges. Bin errors are standard errors of Fourier-mode power, counting each conjugate pair of a real image once. The window can still correlate distinct modes; this is not a complete spectral-covariance treatment.

For each of five time- and detector-position-matched STPSF models (Perrin et al. 2014), the unit-flux point-spread function (PSF) is convolved with 64 independent unit white-noise fields. Applying the same mask, annular mean subtraction and Fourier normalisation gives an expectation spectrum $E_j(k)$. The adopted PSF stamps are 129 pixels across. Gaussian simulation fields probe the second-order PSF/window response; they do not require a Gaussian one-point histogram of the observed residual image.

For each PSF we fit

$$P(k) = P_{0,j} E_j(k) + P_{1,j} \quad (3)$$

by inverse-bin-variance weighted least squares (WLS). The fit interval is $0.04 \leq k \leq 0.25 \text{ pixel}^{-1}$, with a lower bound of ten modes across the shorter crop axis. The coefficient covariance $(X^T W X)^{-1}$ is multiplied by $\max(\chi^2_\nu, 1)$. The adopted amplitudes are medians over the five PSFs; windows beginning at $k_{\min} = 0.01$ and 0.03 test low-frequency sensitivity.

The fluctuation power is $P_{\text{SBF}} = P_0 - P_r$, where P_r is the unresolved globular-cluster and background-galaxy contribution. It is estimated by flux-squared integration of a Gaussian globular-cluster luminosity function and a power-law background population below the source catalogue limit. The adopted globular-cluster width is 1.2 mag. Since Equation 1 has already normalised the variance by the mean galaxy light, P_{SBF} is a fluctuation flux: it must not be divided by the model a second time. For pixel solid angle Ω_{pix} ,

$$\bar{m}_X = -2.5 \log_{10}(P_0 - P_r) - 2.5 \log_{10} \left(\frac{10^6 \Omega_{\text{pix}}}{3631} \right). \quad (4)$$

The two annular magnitudes are combined with their saved normalised inverse-spectral-variance weights, not area weights. F090W is registered and PSF-matched to F150W for the matched-annulus colour measurement. Each band's combined colour uses that band's SBF weights; the two combined colours can therefore differ slightly. Annular magnitude and colour differences remain radial diagnostics, not random-error floors.

2.3. Extinction and calibration

We use Paper IV's A_{090} with $E(B-V) = A_{090}/1.4156$ and $A_{150} = 0.6021 E(B-V)$. Then $\bar{m}_{X,0} = \bar{m}_{X,\text{obs}} - A_X$ and $C_0 = C_{\text{obs}} - (A_{090} - A_{150})$. The absolute calibration points are $\bar{M}_{X,0} =$

$\bar{m}_{X,0} - \mu_{\text{TRGB}}$. Colour first enters at this stage, through a trial calibration $f_X(C_0)$; it never enters Equation 3.

Constant and linear relations are fitted by a Gaussian likelihood with a free non-negative intrinsic scatter σ_{int} . For effective variance

$$V_i = \sigma_{\bar{M},i}^2 + [f'(C_i) \sigma_{C,i}]^2 - 2f'(C_i) \text{Cov}(C_i, \bar{M}_i) + \sigma_{\text{int}}^2, \quad (5)$$

the objective contains both residual-squared/ V_i and $\ln V_i$. Model selection uses AICc and leave-one-galaxy-out (LOO) prediction. Each target is omitted before calibrating its distance. Quadratic, broken-line, logarithmic and exponential laws remain exploratory; they do not replace the primary models solely because a curve looks tighter.

3. Common uncertainty model

An apparent SBF measurement, an absolute calibration point and a predicted distance have different uncertainties. Shared scale terms are correlated between galaxies and are not independent regression weights.

3.1. Image measurement and ring combination

Writing $h = 2.5 / \ln 10$ and MAD for the median absolute deviation, the ring-level spectral error is

$$\sigma_{\text{PS},r} = \max \left[\frac{h \sigma_{P_0,\text{WLS}}}{P_0 - P_r}, 1.4826 \text{ MAD} \{ \bar{m}_r(k_{\min}) \} \right]. \quad (6)$$

The maximum treats two estimates of uncertainty in the same amplitude, rather than adding them as independent physical sources. It is a conservative implementation choice, not Jensen's exact error prescription. Other ring-level contributions are

$$\sigma_{\text{PSF},r} = \frac{h 1.4826 \text{ MAD} \{ P_{0,r,j} \}}{P_{0,r} - P_{r,r}}, \quad (7)$$

$$\sigma_{\text{sky},r} = \frac{h s_{\text{sky}}}{\langle M \rangle_r}, \quad \sigma_{P_r,r} = \frac{h(0.25) P_{r,r}}{P_{0,r} - P_{r,r}}. \quad (8)$$

PSF scatter is measured in power before propagation to magnitudes. Here s_{sky} is the scatter of independent sky estimates. The sky term is a first-order normalisation sensitivity, not a complete sky/model refit. The $0.25P_r$ allowance is assumed, rather than derived from artificial-source completeness tests.

For saved weights $w_r = \sigma_r^{-2} / \sum_s \sigma_s^{-2}$, independent spectral terms combine as $\sum_r w_r^2 \sigma_{\text{PS},r}^2$. Each PSF, sky and P_r term is combined linearly across rings, $\sigma_j = \sum_r w_r \sigma_{j,r}$, because its shift is treated as coherent. The internal image-measurement variance is

$$\sigma_{\bar{m},\text{int}}^2 = \sigma_{\text{PS}}^2 + \sigma_{\text{PSF}}^2 + \sigma_{\text{sky}}^2 + \sigma_{P_r}^2. \quad (9)$$

No annular half-difference is added to this quantity.

3.2. Calibration-point and distance errors

The adopted reddening uncertainty is $\sigma_{A_{090}}^2 = (0.16 A_{090})^2 + 0.005^2$. The individual Paper IV error already contains A_{090} ; we remove that term before restoring the appropriate shared-extinction covariance. Defining $\sigma_{\mu,\text{nr}}^2 = \sigma_{\mu,\text{IV}}^2 - \sigma_{A_{090}}^2$ and $\sigma_{\Delta A} = (1.4156 - 0.6021) \sigma_{E(B-V)}$, we obtain

$$\sigma_{\bar{M}_{150}}^2 = \sigma_{\bar{m}_{150},\text{int}}^2 + \sigma_{\mu,\text{nr}}^2 + \sigma_{\Delta A}^2, \quad (10)$$

$$\sigma_{\bar{M}_{090}}^2 = \sigma_{\bar{m}_{090},\text{int}}^2 + \sigma_{\mu,\text{nr}}^2. \quad (11)$$

The foreground colour–magnitude covariance is $-\sigma_{AA}^2$ in F150W and zero in F090W, where the same A_{090} cancels against the TRGB anchor. These statements concern only foreground extinction, not every shared-image covariance.

The formal photometric colour error comes from the pixel-colour standard error within matched rings, combined as $\sum_r w_r^2 \sigma_{C,r}^2$. Spatially correlated pixels and image-wide photometric systematics are not fully represented. A separate 0.01-mag colour-floor sensitivity test does not reverse the adopted model choice in either band.

With $R_X = A_X/E(B-V)$ and $\sigma_E = \sigma_{E(B-V)}$, a target has $\mu = \bar{m}_{X,\text{obs}} - R_X E(B-V) - f_X(C_0)$. Its measurement and reddening propagation is

$$\sigma_{\text{target}}^2 = \sigma_{m_X,\text{int}}^2 + [f'_X \sigma_{C,\text{phot}}]^2 + [R_X - f'_X(R_{090} - R_{150})]^2 \sigma_E^2. \quad (12)$$

Thus shared reddening is counted once. The predictive error is

$$\sigma_{\mu,\text{int}}^2 = \sigma_{\text{target}}^2 + \sigma_{\text{int}}^2 + \sigma_{\text{cal}}^2, \quad (13)$$

$$\sigma_{\mu,\text{abs}}^2 = \sigma_{\mu,\text{int}}^2 + 0.047^2. \quad (14)$$

The scatter and finite-calibration terms are determined from each 13-galaxy LOO training set; σ_{cal} is the standard deviation of bootstrap predictions (300 realisations for F150W, 350 for F090W). The 0.047-mag common TRGB/NGC 4258 scale (Reid et al. 2019; Chazov et al. 2026) moves all absolute distances together.

For the absolute F150W zero point, common TRGB, flux-scale and PSF-stamp terms are 0.047, 0.0108 and 0.0167 mag, combining to 0.0510 mag. The latter two cancel to first order between apparent and calibrated absolute magnitudes in a same-filter, same-pipeline distance. They are not added again as independent distance errors. We convert moduli using $D = 10^{(\mu - 25)/5}$ Mpc and $\sigma_D \simeq (\ln 10/5) D \sigma_\mu$. Validation residuals $\mu_{\text{SBF}} - \mu_{\text{TRGB}}$ instead include the individual target anchor error with its reddening covariance; the common TRGB zero point cancels. This validation uncertainty, the empirical LOO RMS and the full absolute distance error must not be interchanged.

4. Results in F150W and F090W

4.1. Measurements and model selection

Tables 1 and 2 retain all 14 objects. The tabulated absolute magnitudes are image measurements minus individual TRGB moduli, not values predicted by a colour law. Figure 1 and Table 3 compare both primary models in each band.

The adopted F150W relation is

$$\bar{M}_{150,0} = -3.208 \pm 0.024_{\text{stat}} \pm 0.051_{\text{common}} \text{ mag}, \quad (15)$$

with intrinsic scatter 0.081 mag. The statistical error comes from 2000 galaxy-level bootstrap zero points. The alternative linear slope 0.2315 has an approximate 1σ -equivalent bootstrap interval $[-0.2977, 0.7217]$, including zero. It worsens AICc by 3.14 and the LOO RMS from 0.0925 to 0.0981 mag. Colour is therefore not required for the combined F150W measurements in this sample.

For F090W, the adopted calibration is

$$\bar{M}_{090,0} = (-1.9617 \pm 0.0157_{\text{boot}}) + (1.8890 \pm 0.4424_{\text{boot}})(C_0 - 0.5703), \quad (16)$$

with intrinsic scatter 0.0489 mag. Its linear colour term improves AICc by 9.15 and reduces the LOO RMS by 0.0308 mag. A logarithmic relation changes AICc by only 0.33 and LOO RMS by

0.0011 mag relative to the linear model, insufficient reason to adopt a less direct population law. F150W nonlinear experiments likewise do not improve both model-selection criteria materially.

The paired measurement variances in Figure 2 show how the same image-error prescription contributes in each band. They exclude the population and distance-calibration terms.

4.2. Comparison on the same galaxies

For the eight objects common to this work and Paper III, the F150W constant and linear models give LOO RMS 0.1014 and 0.1103 mag. For Paper III F110W against $g-z$, a linear relation reduces RMS from 0.1114 to 0.0465 mag. Changing the galaxy list therefore does not explain the different colour response.

Figure 3 crosses three SBF bands with both available colour indices. F150W has Pearson $r = 0.28$ against F090W–F150W and $r = 0.21$ against $g-z$ ($p = 0.503$ and 0.611). F110W gives $r = 0.82$ and 0.92 ($p = 0.0127$ and 0.0010). Its relation survives either colour choice, unlike the combined F150W measurements. Passband is therefore an important difference, but population response, distance prescription and reduction choices are not isolated well enough to establish a unique physical explanation.

4.3. Distance recovery and error budgets

The adopted F150W and F090W relations give LOO RMS 0.0925 and 0.0684 mag and biases -0.0010 and -0.0039 mag, respectively (Figure 4). Median internal predictive errors are 0.0865 and 0.0587 mag; including the common TRGB scale gives 0.0984 and 0.0752 mag. Validation-pull RMS values are 1.08 and 1.12. All F150W targets and 13 of 14 F090W targets lie within 2σ ; NGC 4649 is the 2.49σ F090W exception and is retained.

Figures 2 and 5 separate the error in measuring a fluctuation from the error in using it as a distance indicator. Stacked segments add *variances*, in mag^2 , not standard deviations. Small image errors do not imply equally small distance errors: fitted population scatter and finite calibration dominate much of the predictive budget.

4.4. Sensitivity and remaining measurement limitations

Paper IV thick-slice and Paper-III-style cluster anchors give F150W constant-model LOO RMS of approximately 0.093 mag, close to the primary result; cluster tests separate depth from shared mean-distance errors. Fixed quality cuts do not justify removing objects or restore a useful F150W colour law. The Virgo–Fornax step remains diagnostic, not a detection.

Radial behaviour is not identical. F150W inner and outer annuli have slopes 1.01 and 0.11 and intrinsic scatters 0.050 and 0.090 mag. F090W slopes are 2.62 and 1.18. Both rings are retained to match the published angular geometry, not because they are independent or equally stable. A null combined F150W colour result is not a claim of no local population gradient.

Internal PSF-size tests give maximum 129–257-pixel shifts of 0.0167 mag in F150W and 0.0128 mag in F090W; an empirical isolated-star validation is still missing. The F150W $0-2P_r$ test changes weighted magnitudes by at most 0.0053 mag, and the special 1.35-mag globular-cluster widths for NGC 1399 and NGC 4472 change them by at most 0.00047 mag. This bounds the tested correction, not catalogue incompleteness itself.

Relative to the former raw-residual winsorisation, the adopted normalised-space F150W branch shifts the combined magnitude by

Table 1. F150W extinction-corrected measurements. $C_{090-150,0}$ is the matched colour; $\bar{m}_{150,0}$ is apparent SBF and $\bar{M}_{150,0} = \bar{m}_{150,0} - \mu_{\text{TRGB}}$. The individual $\sigma_{\bar{M}}$ follows Equation 11, excluding common zero points. $|\Delta\bar{m}|_{\text{ann}}$ is the full absolute annular difference, not an error term; $\Delta\mu_{\text{LOO,const}}$ is the colour-independent validation residual. HQ IV denotes Paper IV’s quality flag and does not select the primary sample. All magnitudes are in mag.

Galaxy	Environment	$C_{090-150,0}$	$\bar{m}_{150,0}$	$\bar{M}_{150,0}$	$\sigma_{\bar{M}}$	$ \Delta\bar{m} _{\text{ann}}$	μ_{TRGB}	$\Delta\mu_{\text{LOO,const}}$	HQ IV
		(mag)	(mag)	(mag)	(mag)	(mag)	(mag)	(mag)	
NGC 1380	Fornax	0.510	28.087	-3.321	0.029	0.099	31.408	-0.122	+
NGC 1399	Fornax	0.569	28.157	-3.341	0.034	0.047	31.498	-0.143	–
NGC 1404	Fornax	0.559	28.177	-3.189	0.028	0.110	31.366	+0.020	+
NGC 1549	Other	0.533	27.930	-3.301	0.029	0.133	31.231	-0.101	–
NGC 3379	Other	0.558	27.012	-3.066	0.030	0.269	30.078	+0.152	+
NGC 4374	Virgo region	0.566	27.947	-3.165	0.031	0.094	31.112	+0.046	+
NGC 4406	Virgo region	0.616	27.849	-3.247	0.030	0.097	31.096	-0.042	+
NGC 4472	Virgo region	0.624	27.713	-3.303	0.035	0.029	31.016	-0.103	–
NGC 4486	Virgo region	0.652	27.808	-3.238	0.055	0.054	31.046	-0.033	–
NGC 4552	Virgo region	0.570	27.850	-3.073	0.030	0.175	30.923	+0.146	+
NGC 4621	Virgo region	0.566	27.688	-3.133	0.028	0.173	30.821	+0.081	+
NGC 4636	Virgo region	0.604	27.891	-3.179	0.031	0.110	31.070	+0.031	+
NGC 4649	Virgo region	0.623	27.873	-3.136	0.030	0.043	31.009	+0.077	–
NGC 4697	Virgo region	0.512	27.095	-3.229	0.032	0.173	30.324	-0.023	+

Table 2. F090W extinction-corrected measurements, with the same object selection and definitions as Table 1. $\sigma_{\bar{m}}$ is the internal image-measurement error from Equation 9; $\sigma_{\bar{M}}$ additionally contains the individual TRGB term with the shared A_{090} cancellation. The colour uses the F090W annular weights, rather than the F150W weights.

Galaxy	Env.	$C_{090-150,0}$	$\bar{m}_{090,0}$	$\sigma_{\bar{m}}$	$\bar{M}_{090,0}$	$\sigma_{\bar{M}}$	$ \Delta\bar{m} _{\text{ann}}$	HQ IV
		(mag)	(mag)	(mag)	(mag)	(mag)	(mag)	
NGC 1380	Fornax	0.511	29.328	0.020	-2.080	0.032	0.039	+
NGC 1399	Fornax	0.569	29.616	0.023	-1.882	0.034	0.079	–
NGC 1404	Fornax	0.565	29.414	0.016	-1.952	0.029	0.006	+
NGC 1549	Other	0.539	29.214	0.027	-2.017	0.037	0.045	–
NGC 3379	Other	0.570	28.156	0.017	-1.922	0.032	0.156	+
NGC 4374	Virgo region	0.566	29.163	0.023	-1.949	0.034	0.029	+
NGC 4406	Virgo region	0.615	29.130	0.024	-1.966	0.034	0.040	+
NGC 4472	Virgo region	0.626	29.128	0.016	-1.888	0.031	0.018	–
NGC 4486	Virgo region	0.652	29.199	0.057	-1.847	0.062	0.056	–
NGC 4552	Virgo region	0.574	29.025	0.019	-1.898	0.033	0.088	+
NGC 4621	Virgo region	0.571	28.806	0.019	-2.015	0.031	0.075	+
NGC 4636	Virgo region	0.603	29.097	0.024	-1.973	0.034	0.041	+
NGC 4649	Virgo region	0.622	29.254	0.013	-1.755	0.027	0.045	–
NGC 4697	Virgo region	0.517	28.198	0.016	-2.126	0.031	0.057	+

Table 3. Constant and linear calibrations for the same 14 galaxies. The linear form is $a + b(C_0 - C_p)$, with $C_p = 0.5675$ mag for F150W and 0.5703 mag for F090W. Lower AICc and LOO RMS are preferred; the adopted models are marked by an asterisk. a , σ_{int} and LOO RMS are in mag.

Band	Model	a	b	σ_{int}	AICc	LOO RMS
F150W	Constant*	-3.2078	0	0.0805	-23.75	0.0925
F150W	Linear	-3.2094	0.2315	0.0797	-20.61	0.0981
F090W	Constant	-1.949	0	0.086	-21.84	0.0992
F090W	Linear*	-1.9617	1.8890	0.0489	-30.98	0.0684

a median -0.02954 mag and up to 0.08798 mag in absolute value. The order of operations is therefore a material methodological choice, not a negligible correction.

The adopted noise model is imperfect. Median spectral reduced χ^2 values are 19.94 (F150W) and 4.48 (F090W), even after conjugate mode counting is corrected. All adopted F150W ring fits have $P_1 < 0$; the intercept approximates a correlated-noise continuum

locally and is not negative physical noise power. An empirical $N(k)$ extension is nearly degenerate with $E(k)$ (correlations 0.982–0.997) and does not provide a stable alternative. Archived synthetic FFT and winsorisation checks constrain normalisation, not full reduction systematics; they predate the bin-error correction and do not validate its uncertainty coverage. Detailed diagnostics are in the Technical notes.

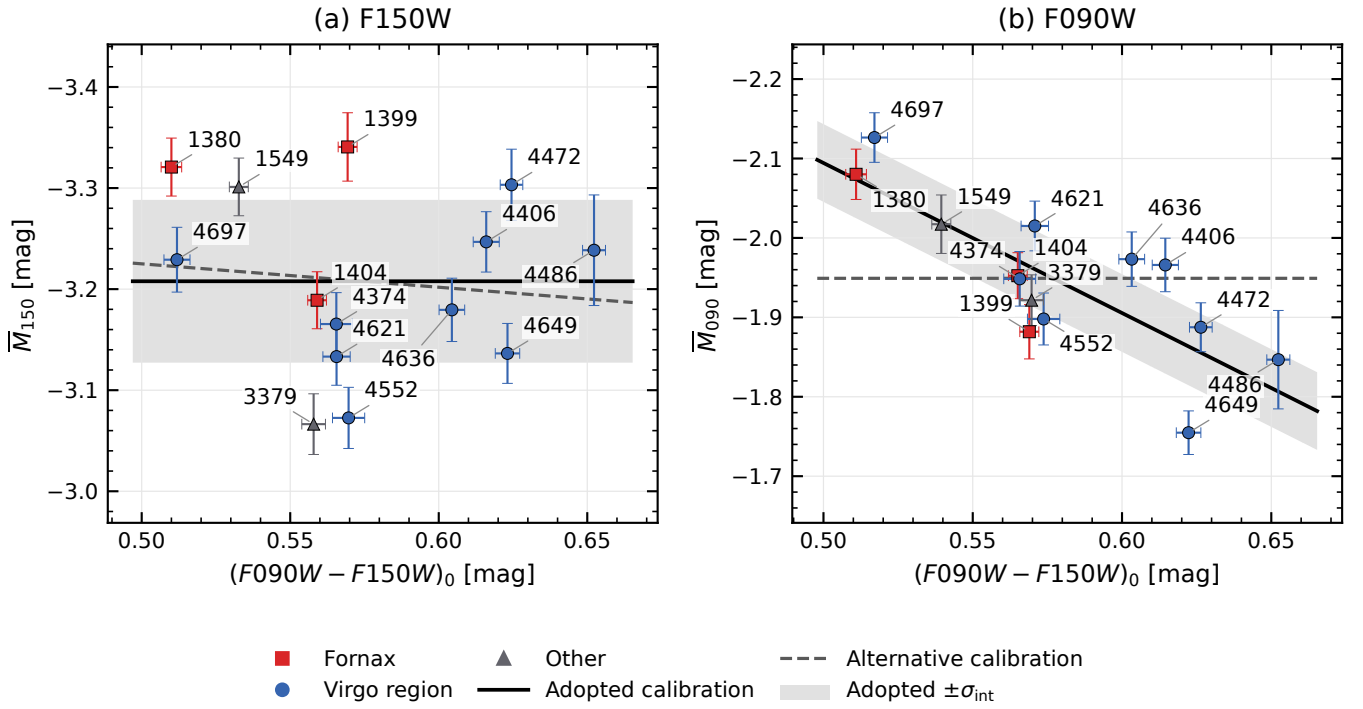


Figure 1. Joint calibration comparison. The solid black line is the adopted constant in F150W (left) and linear colour relation in F090W (right); the dashed line is the alternative primary model. Each grey strip shows only the adopted $\pm\sigma_{\text{int}}$, not a confidence interval on the fitted mean or a full distance error. Red squares denote Fornax, blue circles the broad Virgo region, and grey triangles other environments; labels are NGC numbers. All 14 galaxies enter each fit. Bars are individual 1σ colour and absolute-SBF errors, excluding common zero points.

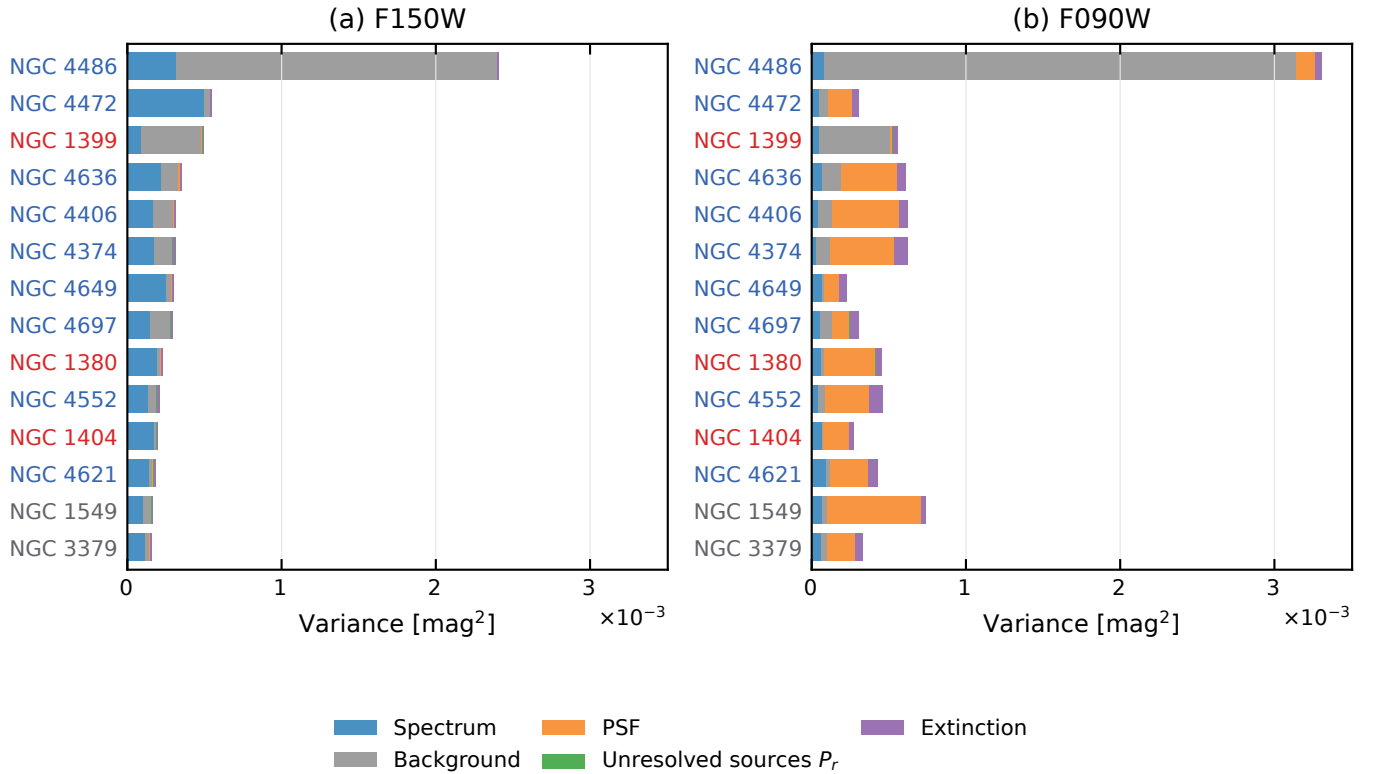


Figure 2. Paired apparent-SBF measurement budgets. Each horizontal segment is a contribution σ_i^2 ; the total includes foreground extinction in addition to the internal terms of Equation 9. Both panels use the same galaxy order. These are not full distance uncertainties.

Same-galaxy colour comparison across three SBF bands

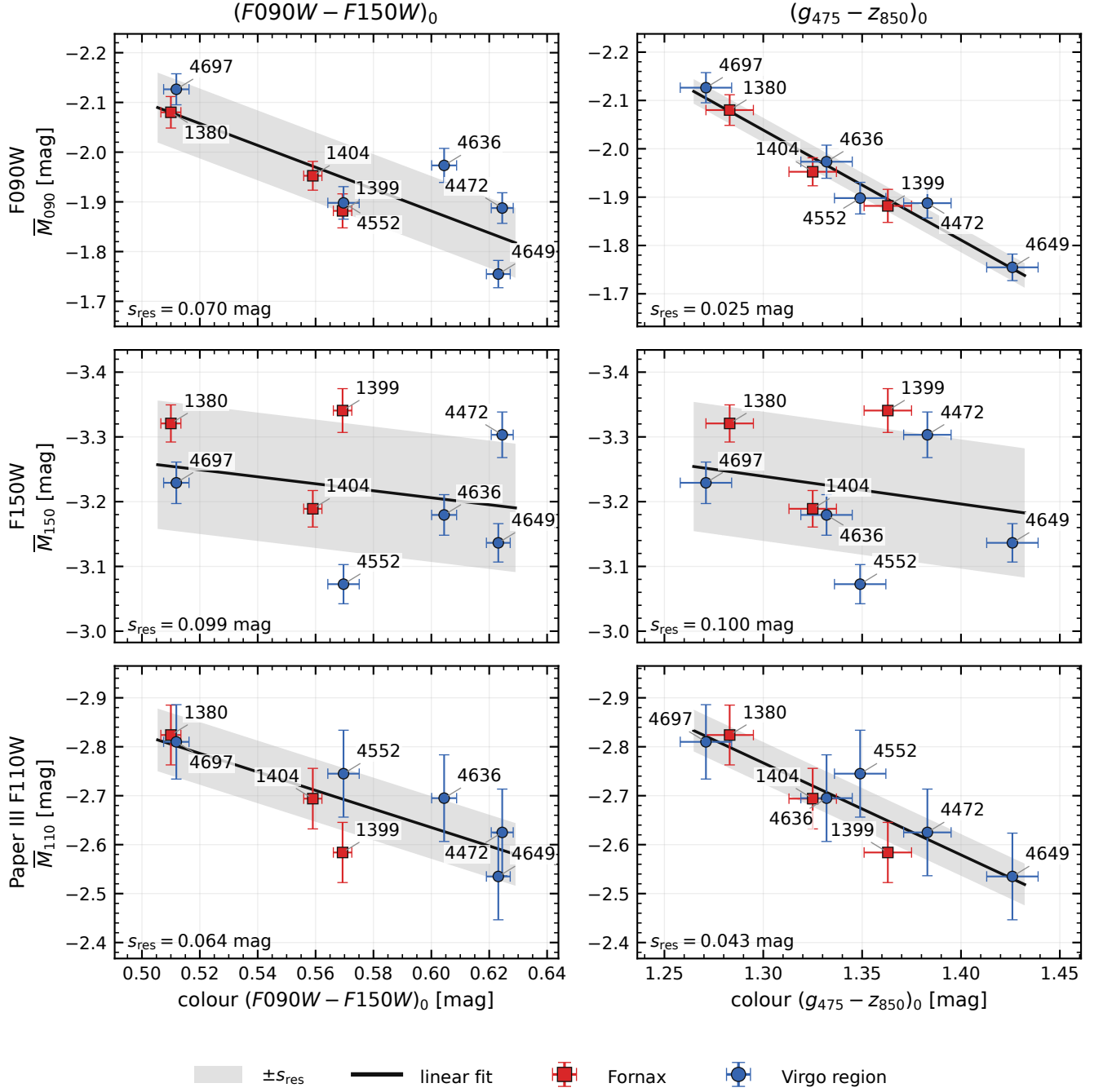


Figure 3. Same-eight-galaxy comparison across SBF bands (rows) and colour indices (columns). Symbols identify environment as in Figure 1; all points carry NGC labels and individual error bars. The ordinary least-squares lines are diagnostic, not the adopted calibrations. Each single strip is $\pm s_{\text{res}}$, with $s_{\text{res}}^2 = \sum_i (y_i - \hat{y}_i)^2 / (8 - 2)$: observed residual scatter including measurement noise, not fitted intrinsic scatter or coefficient uncertainty.

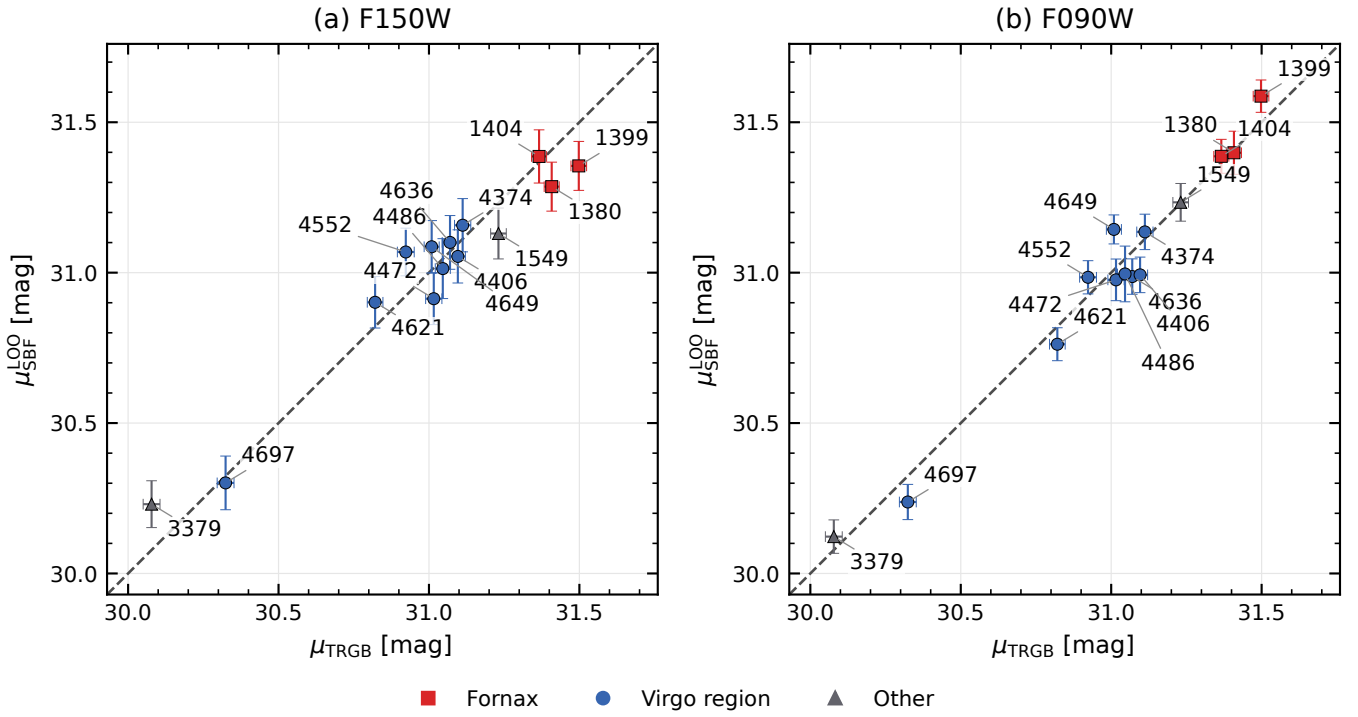


Figure 4. LOO distance recovery for the adopted F150W constant (left) and F090W linear calibration (right). Each target is omitted from its calibration; dashed lines mark equality. Horizontal bars are individual TRGB errors and vertical bars are internal SBF predictive errors. The common absolute scale is excluded. Environment symbols and NGC labels follow Figure 1. This is internal cross-validation, not an independent external test.

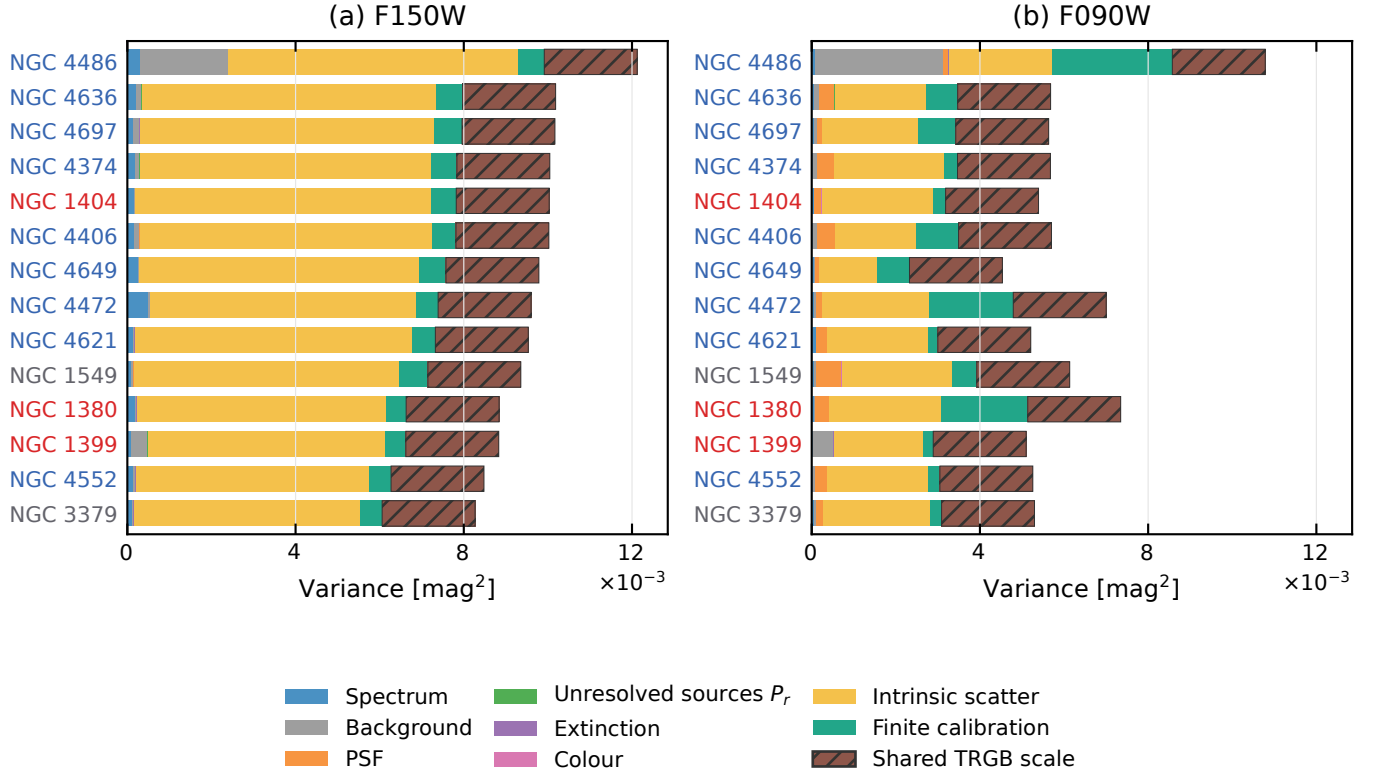


Figure 5. Paired full LOO distance-error budgets. Segments show the individual measurement, colour/reddening propagation, intrinsic scatter, finite calibration and shared TRGB-scale variances. Colour contributes only for F090W. The common scale is displayed inside each total but is correlated between galaxies, not an independent per-object error.

5. Adopted distances

Table 4 is the final distance comparison. Every error is a full absolute 1σ uncertainty in Mpc. TRGB IV and both JWST columns include the common 0.047-mag TRGB scale. The Paper III F110W reconstruction includes measurement and colour errors, slope uncertainty, 0.060-mag population scatter and its 0.063-mag common scale; its 0.016-mag transfer term is already included in that scale. The separate Jensen et al. (2015) F160W reconstruction includes measurement, colour and coefficient errors, 0.114-mag scatter and the 0.10-mag common Cepheid zero point (Jensen et al. 2015, 2021, 2025). Common terms must be cancelled or represented through covariance when comparing columns; the printed errors are appropriate to quoting each absolute distance in isolation.

Table 4. Final distances in Mpc with full absolute 1σ errors. TRGB IV: individual thin-slice anchors; Jensen III: HST F110W; Jensen 2015: HST F160W. This work: colour-independent F150W and colour-dependent F090W LOO distances. Ellipses indicate absent comparison measurements. Shared scale terms remain correlated.

Galaxy	TRGB IV	Jensen III	Jensen 2015	This work	
		F110W	F160W	F150W	F090W
NGC 1380	19.12 ± 0.47	18.98 ± 0.81	20.12 ± 1.51	18.08 ± 0.78	19.04 ± 0.75
NGC 1399	19.93 ± 0.49	19.80 ± 0.85	20.30 ± 1.51	18.66 ± 0.81	20.77 ± 0.68
NGC 1404	18.76 ± 0.46	19.44 ± 0.83	19.84 ± 1.53	18.94 ± 0.87	18.94 ± 0.64
NGC 1549	17.63 ± 0.44	...	...	16.83 ± 0.75	17.65 ± 0.64
NGC 3379	10.37 ± 0.26	...	...	11.12 ± 0.47	10.58 ± 0.35
NGC 4374	16.69 ± 0.42	...	...	17.04 ± 0.79	16.87 ± 0.59
NGC 4406	16.57 ± 0.41	...	...	16.25 ± 0.75	15.80 ± 0.55
NGC 4472	15.97 ± 0.40	16.11 ± 0.69	16.19 ± 1.23	15.23 ± 0.69	15.68 ± 0.60
NGC 4486	16.19 ± 0.40	...	...	15.95 ± 0.81	15.82 ± 0.76
NGC 4552	15.30 ± 0.39	15.70 ± 0.67	...	16.36 ± 0.69	15.74 ± 0.53
NGC 4621	14.59 ± 0.36	...	...	15.15 ± 0.68	14.20 ± 0.47
NGC 4636	16.37 ± 0.40	16.30 ± 0.70	...	16.60 ± 0.77	15.75 ± 0.55
NGC 4649	15.91 ± 0.39	16.19 ± 0.71	15.86 ± 1.23	16.49 ± 0.75	16.93 ± 0.53
NGC 4697	11.61 ± 0.29	11.66 ± 0.50	...	11.49 ± 0.53	11.16 ± 0.39

6. Discussion

F090W provides the tighter internal distance calibration on these 14 galaxies. F150W remains competitive without colour, while a common population law for both filters is not supported. The comparison with the same eight Paper III galaxies shows that the difference is not simply a change in sample membership or choice of colour index. However, the significant inner-annulus F150W trend and the limited colour range prevent a universal assertion that F150W is colour independent.

Under the full-error convention, median uncertainties are 0.70 Mpc for the eight Paper III F110W overlaps, 0.75 Mpc for F150W and 0.57 Mpc for F090W. These medians describe different overlap sets and distances; they are not a controlled demonstration of universal instrumental superiority. The five F160W comparisons give an SBF-modulus offset -0.119 ± 0.133 mag using joint covariance, which does not establish a significant absolute-scale mismatch. The Technical notes give the conditional covariance assumptions.

LOO tests common reductions and a shared anchor scale, so it cannot measure external accuracy or uncover an offset common to every calibrator. Residual structure, correlated noise, empirical PSF normalisation and formal colour errors remain important limitations. No internal-dust or object-specific SBF K -correction has been applied. All targets have $z < 0.007$, but an ordinary integrated-light K estimate is not a bound on the luminosity-moment-weighted SBF correction (Cervino et al. 2008). No empirical internal-dust upper limit is claimed. Cosmological flux dimming is already represented through luminosity distance; an extra Tolman term is not added (Hogg 1999). An independent sample and more complete noise/population treatment are needed before claiming external precision at the internal LOO level.

7. Conclusions

The same image-measurement procedure yields two useful but distinct TRGB-anchored calibrations. All 14 targets are retained. A constant $\bar{M}_{150,0} = -3.208$ mag gives F150W LOO RMS 0.0925 mag; colour does not improve that prediction. F090W instead favours a linear colour slope 1.889 ± 0.442 and reaches 0.0684-mag LOO RMS. The median full absolute errors are 0.0984 and 0.0752 mag, respectively. These are internal results with explicit common-scale and unresolved measurement systematics, not an independent validation of the absolute distance scale.

8. Data and code availability

The calibrated JWST observations are public in MAST under GO-3055, doi:10.17909/z9ch-wk24. Derived tables, image products, analysis code and the accompanying Technical notes are retained in the version-controlled project. The implementation uses NumPy, SciPy, Astropy, Photutils, Matplotlib and STPSF (Harris et al. 2020; Virtanen et al. 2020; Astropy Collaboration et al. 2022; Hunter 2007; Perrin et al. 2014); exact processing versions and input-header contexts are recorded in the Technical notes. The 28 input mosaics have mixed archive pipeline/CRDS contexts rather than a uniform local reprocessing. A permanent public code/data release has not yet been assigned; this statement does not claim a local repository is public.

References

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, *ApJ*, 935, 167, doi:10.3847/1538-4357/ac7c74.

- Cerviño, M., Luridiana, V., & Jamet, L. 2008, *On surface brightness fluctuations: probabilistic and statistical bases. I. Stellar population and theoretical SBF*, arXiv:0809.4491, doi:10.1051/0004-6361:20077515.
- Chazov, M. I., Makarov, D. I., Tully, R. B., et al. 2026, *The TRGB–SBF Project. IV. A Color Calibration of the TRGB in the JWST F090W+F150W Filters*, arXiv:2603.11160v1.
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, *Nature*, 585, 357, doi:10.1038/s41586-020-2649-2.
- Hogg, D. W. 1999, *Distance measures in cosmology*, arXiv:astro-ph/9905116.
- Hunter, J. D. 2007, *Computing in Science & Engineering*, 9, 90, doi:10.1109/MCSE.2007.55.
- Jensen, J. B., Tonry, J. L., Barris, B. J., et al. 2003, *ApJ*, 583, 712, doi:10.1086/345430.
- Jensen, J. B., Blakeslee, J. P., Ma, C.-P., et al. 2015, *ApJ*, 808, 91, arXiv:1505.00400.
- Jensen, J. B., Blakeslee, J. P., Ma, C.-P., et al. 2021, *ApJS*, 255, 21, doi:10.3847/1538-4365/ac01e7.
- Jensen, J. B., Blakeslee, J. P., Cantiello, M., et al. 2025, *The TRGB–SBF Project. III. Refining the HST Surface Brightness Fluctuation Distance Scale Calibration with JWST*, *ApJ*, 987, 87, doi:10.3847/1538-4357/addfd6.
- Perrin, M. D., Sivaramakrishnan, A., Lajoie, C.-P., et al. 2014, *Proc. SPIE*, 9143, 91433X, doi:10.1117/12.2056689.
- Reid, M. J., Pesce, D. W., & Riess, A. G. 2019, *ApJL*, 886, L27, doi:10.3847/2041-8213/ab552d.
- Tonry, J. L., & Schneider, D. P. 1988, *AJ*, 96, 807.
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, *Nature Methods*, 17, 261, doi:10.1038/s41592-019-0686-2.